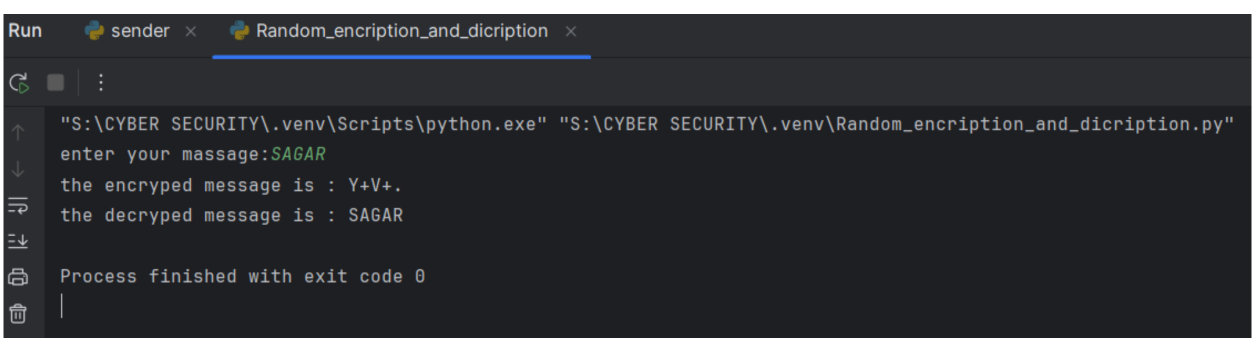
1. Cryptography (Random Encryption)

```
import random
message=input("enter your massage:")
keys=['0', '1', '2', '3', '4', '5', '6', '7', '8', '9', 'a', 'b', 'c', 'd', 'e',
    'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't',
    'u', 'v', 'w', 'x', 'y', 'z', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I',
    'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V', 'W', 'X',
    'Y', 'Z', '!', '"', '#', '$', '%', '&', "'", '(', ')', '*', '+', ',', '-',
    '.', '/', ':', ';', '<', '=', '>', '?', '@', '[', '\\', ']', '^', '_', '`',
    '{', '|', '}', '~', ' ', '\t', '\n', '\r', '\x0b', '\x0c']
val=keys[::-1]
random_val = keys[:]
random.shuffle(random_val)
#print(val)
e_msg=dict(zip(keys,random_val))
#print(e_msg)
enc_msg="".join(e_msg[words]for words in message)
print("the encryped message is :",enc_msg)

d_msg=dict(zip(random_val,keys))
dnc_msg="".join(d_msg[words]for words in enc_msg)
print("the decryped message is :",dnc_msg)
```

-: OUTPUT :-



2. Server Hack (Random Encryption)

SENDER.PY

```
import socket
import random
import string
key_chars = list(string.printable[:94])
random.shuffle(key_chars)
key = "".join(key_chars)

original_chars = string.printable[:94]
encrypter = dict(zip(original_chars, key))

message = input("Enter your message: ")
encrypted_message = "".join([encrypter[char] if char in encrypter else char for char in message])

print(f"Encryption Key (Share with Receiver): {key}")  # Display for debugging
print(f"Encrypted Message: {encrypted_message}")

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.bind((socket.gethostname(), 3125))
s.listen(5)

print("Sender is waiting for a receiver...")

while True:
    client, address = s.accept()
    print(f"Connection from {address} established.")

    client.send(bytes(key, "utf-8"))
    client.send(bytes(encrypted_message, "utf-8"))

    client.close()
```

-: OUTPUT :-

```
"S:\CYBER SECURITY\.venv\Scripts\python.exe" "S:\CYBER SECURITY\.venv\sender.py"
Enter your message: SAGAR
Encryption Key (Share with Receiver): lsmK5St/nqx)g}8|<1_~Ly0$v6{h\2H4AWP@UVTQrbD:j+JoZ3B-CG7(*#;Xf'N,R9pi0c`"e=]Ew!k[z%MFu.Y&a?I>d^
Encrypted Message: 7UDUG
Sender is waiting for a receiver...
```

RECIVER.PY

```
import socket

s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((socket.gethostname(), 3125))

key = s.recv(94).decode("utf-8")

message = s.recv(1000).decode("utf-8")

original_chars =
"0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ
!#$%&'()*+,-./:;<=>?@[\]^_`{|}~"
decrypter = dict(zip(key, original_chars))

decrypted_message = "".join([decrypter[char] if char in decrypter else char for char in message])

print("Decrypted Message:", decrypted_message)

s.close()
```

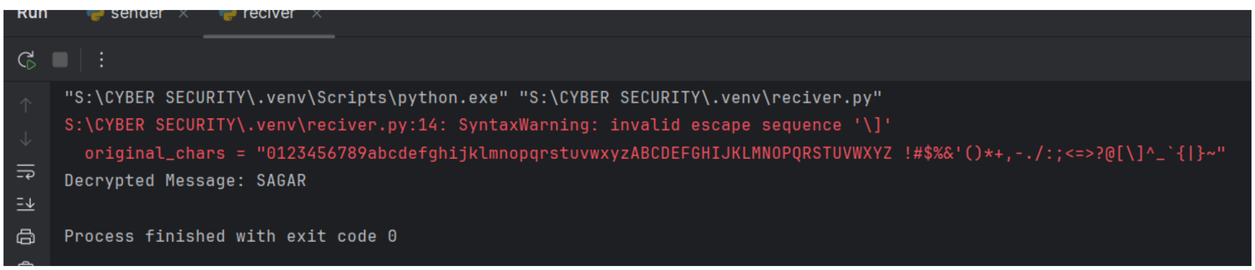
-: OUTPUT :-

```
"S:\CYBER SECURITY\.venv\Scripts\python.exe" "S:\CYBER SECURITY\.venv\reciver.py"
S:\CYBER SECURITY\.venv\reciver.py:14: SyntaxWarning: invalid escape sequence '\]'
  original_chars = "0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ !#$%&'()*+,-./:;<=>?@[\]^_`{|}~"
Decrypted Message: SAGAR

Process finished with exit code 0
```

HACKER.PY

```
import socket


s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((socket.gethostname(), 3125))


key = s.recv(94).decode("utf-8")


encrypted_message = s.recv(1000).decode("utf-8")

print("Hacker intercepted:")
print("Encryption Key:", key)
print("Encrypted Message:", encrypted_message)


print("\nHacker attempting decryption (without mapping)...")
brute_force_attempt = encrypted_message[::-1]
print("Brute-force Attempt:", brute_force_attempt)

s.close()
```

-: OUTPUT :-

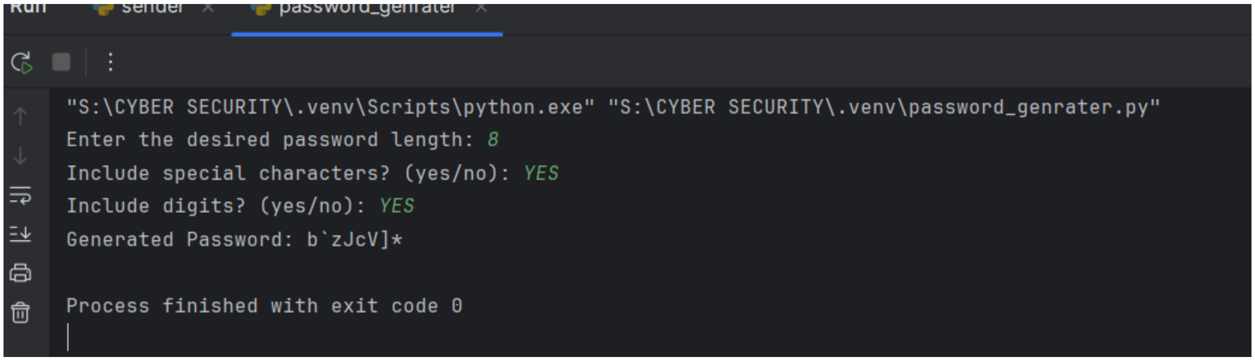


```
"S:\CYBER SECURITY\.venv\Scripts\python.exe" "S:\CYBER SECURITY\.venv\reciver.py"
S:\CYBER SECURITY\.venv\reciver.py:14: SyntaxWarning: invalid escape sequence '\]'
  original_chars = "0123456789abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ !#$%&'()*+,-./:;<=>?@[\]^_`{|}~"
Decrypted Message: SAGAR

Process finished with exit code 0
```

3. Secured Password Generator

```
import secrets
import string


def generate_password(length, include_special=True, include_digits=True):
    char_set = string.ascii_letters

    if include_special:
        char_set += string.punctuation

    if include_digits:
        char_set += string.digits


    password = ''.join(secrets.choice(char_set) for _ in range(length))

    return password



length = int(input("Enter the desired password length: "))

include_special = input("Include special characters? (yes/no): ").strip().lower() == 'yes'
include_digits = input("Include digits? (yes/no): ").strip().lower() == 'yes'


password = generate_password(length, include_special, include_digits)
print("Generated Password:", password)
```

-: OUTPUT :-



Sentiment.py

```
import nltk
from nltk.sentiment import SentimentIntensityAnalyzer

nltk.download('vader_lexicon')
sia = SentimentIntensityAnalyzer()

def analyze_sentiment(text):
    score=sia.polarity_scores(text)
    if score['compound'] >= 0.05:
        return "Positive"
    elif score['compound'] <= -0.05:
        return "Negative"
    else:
        return "Neutral"
text = input("Enter message:")
result = analyze_sentiment(text)
print(result)
print(result)
```

-: OUTPUT :-

Run sender × sentiment ×

```
"S:\CYBER SECURITY\.venv\Scripts\python.exe" "S:\CYBER SECURITY\.venv\sentiment.py"
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\ssaga\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
Enter message:hello malli
Neutral
Neutral
```